

Short course on “Regression Analysis”

Lecture 1: Introduction to Simple and Multiple LR

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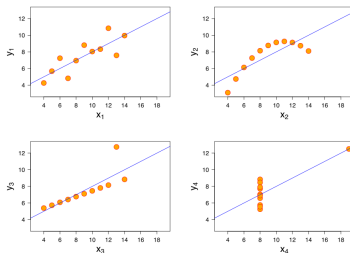
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Linear regression is a **fundamental** statistical method used to

- model the relationship between two or more variables
- by attempting to find the best-fitting straight line that predicts the value of one variable based on the value of one or more other variables



Obviously, we CANNOT make causal conclusions by the use of linear regression, still it is widely used.

Why Use Linear Regression?

■ Simplicity and Interpretability:

- Easy to understand and implement.
- Clear interpretation of coefficients.

■ Efficiency:

- Computationally inexpensive and quick to train.

■ Baseline Model:

- Serves as a benchmark for comparing complex models.
- Helps in evaluating whether added complexity is justified.

■ Foundation for More Complex Models:

- Basis for extensions like multiple linear regression, ridge, and lasso regression.
- Essential for understanding advanced machine learning models.

■ Versatility:

- Applicable across various fields.
- Works well with linearly separable data.

■ Ease of Implementation:

- Supported by many tools and libraries (e.g., scikit-learn, R).
- Requires fewer data assumptions than complex models.

Observational vs. experimental data

Observational data Data arising from non-experimental, non-designed data. They do not control the explanatory variable. Anything that we do on observational data is not causal. So, we can't make any cause-and-effect statements, that is, you can't say *xyz* happened because of *abc*. Typically, the data that is used in regression analysis is observational.

Experimental data If one does a controlled experiment, then the regression on that data could be causal. A controlled experiment is where you offer a "treatment" and "control" to a randomized set of subjects.

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- Company officials wished to study the relationship between age of employee and the number of illness days last year.

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- An insurance company wished to study the relationship between productivity of its analysts in processing claims and length of training.

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Regression analysis

Used for explaining or modeling the relationship between a single variable Y , called the **response, output or dependent variable**; and one or more **predictor, input, independent or explanatory variables**, $X_1, \dots, X_p$. So, we are looking for relationships like

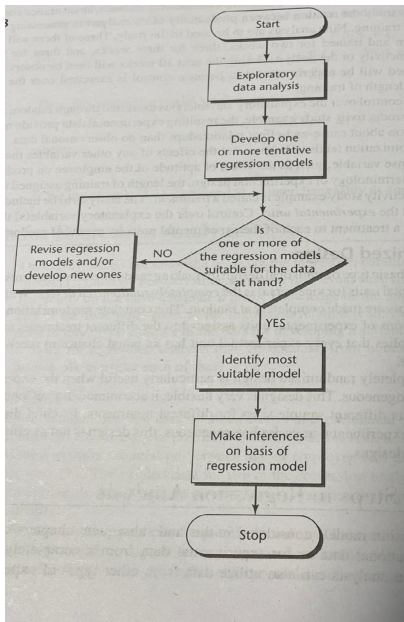
$$Y = \beta_0 + X_1\beta_1 + \dots + X_p\beta_p + \epsilon$$

When $p = 1$, it is called **simple regression** but when $p > 1$ it is called **multiple regression or sometimes multivariate regression**.

When there is more than one Y , then it is called **multivariate multiple regression**. separate regressions on each Y .

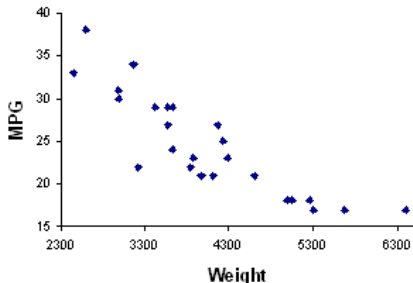
The **response** must be a continuous variable, but the **explanatory** variables can be continuous, discrete or categorical, although we leave the handling of categorical explanatory variables to later

Overview of steps in regression analysis



Example 1: Car weight vs. MPG

- What is the relationship between the weight of a vehicle and the number of miles it travels per gallon of gasoline? To answer this question, a random sample of 25 vehicles was taken. The weight (in pounds) and the miles per gallon (MPG) was recorded for each car.
- Is this observational or experimental data?
- Which variable is the explanatory and which is the response?
 - Explanatory:
 - Response:



Simple Linear Regression Model

The Simple Linear Regression Model is given as:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n$$

- y_i : dependent variable or response variable.
- x_i : independent variable or explanatory variable.
- ε_i : the error term or noise term in order to explain the discrepancy.
 - Assume mean 0 and variance σ^2 , and iid (independent and identically distributed). Conditions: $\mathbb{E}\varepsilon_i = 0$ and $\text{Var } \varepsilon_i = \sigma^2$;
 - Something in the response variable y_i that cannot be completely determined by the explanatory random variable x_i ;
 - For example: for any car that has the same weight, the MPG cannot be completely the same due to
 - road conditions, driver's driving habits, gasoline qualities;
 - all these factors contribute to MPG, whose impact are sort of unpredictable, can be put into the error term ε_i ;
- β_0, β_1 and σ^2 : unknown parameters, need to be estimated

Simple Linear Regression Model-continued

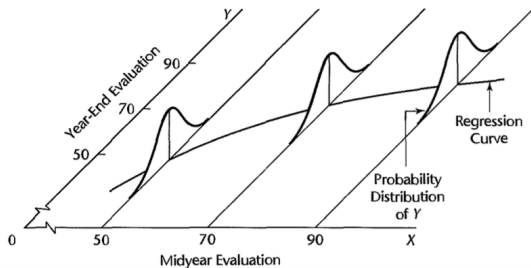
The Simple Linear Regression Model is given as:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n$$

- $E(y_i|x_i) = E(\beta_0 + \beta_1 x_i + \varepsilon_i) = \beta_0 + \beta_1 x_i + E(\varepsilon_i) = \beta_0 + \beta_1 x_i$
- $V(y_i|x_i) = V(\beta_0 + \beta_1 x_i + \varepsilon_i) = V(\varepsilon_i) = \sigma^2$
- β_0 is the intercept term and is fixed but unknown. Interpreted as the expected value of Y when X is zero (needs to be estimated!)
- β_1 is the slope term and is fixed but unknown. Interpreted as the expected increase in the rate of the value of Y when the value of X increases one unit (needs to be estimated!)
- The explanatory random variable x_i is usually observable and thus considered fixed!

Probability Distribution for each x value

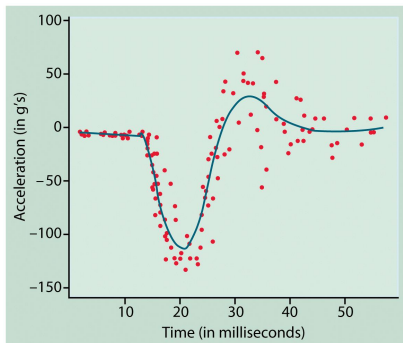
For each value of X , there is a postulated probability distribution for Y . The means of these probability distributions vary in some systematic fashion with X . Say, a company wants to find out a relationship between mid-year performance evaluations and year-end evaluations. In addition to the assumptions on the last page, we also assume that the errors are normally distributed (for the following illustration).



Is the association between two variables really linear?

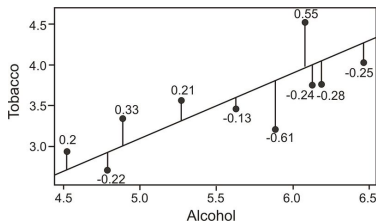
Are the error terms necessarily normal?

- Not always true;
- Nonlinear regression models are also available to deal with that.



Least Square Estimator

- We attempt to find an estimator of β_0 and β_1 , such that y_i 's are overall "close" to the fitted line;
- Define the fitted line as $\hat{y}_i \equiv \hat{\beta}_0 + \hat{\beta}_1 x_i$ and the residual, $e_i = y_i - \hat{y}_i$;



- We define the sum of squared errors (or **residual sum of squares**) to be

$$SSE(\text{RSS}) \equiv \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2$$

- We find a pair of $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize this sum of squares. The resulting estimators are called **least square estimators (LSE)**, or **ordinary least square (OLS)** estimators.

Least Squares Estimate

The SSE measures the distance between the observed response and predicted response values.

By optimization, we know that the values of $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimizing SSE must satisfy the following equations (called normal equations)

$$\begin{cases} \sum y_i = n\beta_0 + \beta_1 \sum x_i \\ \sum x_i y_i = \beta_0 \sum x_i + \beta_1 \sum x_i^2. \end{cases} \quad (1)$$

Solving Equation (1) yields

$$\begin{cases} \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} \\ \hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2} \end{cases}$$

Moreover since $\sum_{i=1}^n \bar{y}(x_i - \bar{x}) = 0$, the estimator for β_1 can also be written as

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \bar{y}}{\sum_{i=1}^n x_i^2 - \sum_{i=1}^n x_i \bar{x}} = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Notations

- Define $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$;
- Define $S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$;
- Finally, the Ordinary Least square estimator for the simple linear regression model are:

$$\begin{cases} \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x}; \\ \hat{\beta}_1 &= \frac{S_{xy}}{S_{xx}}. \end{cases}$$

- The predicted or fitted simple linear regression equation is:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$$

- $\hat{\beta}_0$ and $\hat{\beta}_1$ are the estimates of β_0 , and β_1
- $\hat{Y}$ is the predicted value of Y by inserting a value of X into the prediction equation
- The predicted value $\hat{Y}$
 - estimates the mean of the sub-population of the values of response for a specific value of the predictor?

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 - predict the value of response for a particular case for a specific value of the predictor No

Use the lm() of R to fit the regression model

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```
model1<-lm(KWH~AC,data=kwhac)
summary(model1)
```

Call:

```
lm(formula = KWH ~ AC, data = kwhac)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-26.8976	-6.9441	-0.8627	11.4435	20.7497

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	27.851	7.807	3.568	0.00205	**
AC	5.341	1.031	5.182	5.3e-05	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14.45 on 19 degrees of freedom

Multiple R-squared: 0.5856, Adjusted R-squared: 0.5638

F-statistic: 26.85 on 1 and 19 DF, p-value: 5.301e-05

Residuals

Definition 1

Residual for the i th subject is defined to be the difference between the observed y_i and the predicted $\hat{y}_i$. i.e. $e_i = y_i - \hat{Y}_i$

Could you tell me the differences of the following two terms?

■ ε_i

■ e_i

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- e_i

Good Properties of the residuals:

- $\sum_{i=1}^n e_i = 0$

- $\sum_{i=1}^n e_i^2$ is a minimum

- $\sum_{i=1}^n y_i = \sum_{i=1}^n \hat{y}_i$. In other words, the mean of the fitted values $\hat{Y}_i$ is the same as the mean of the observed values y_i

- $\sum_{i=1}^n x_i e_i = 0$

- $\sum_{i=1}^n \hat{y}_i e_i = 0$

- the fitted regression line always goes through the point $(\bar{X}, \bar{Y})$

Point Estimator of σ^2

Recall that the variance of each observation Y_i for the regression model is σ^2 , the same for each error term.

Recall from “Applied Statistics” that the variance can be estimated unbiasedly by the sample variance with an appropriate divisor.

Residual Sum of Squares, SSE, is defined as

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n e_i^2.$$

SSE has $n - 2$ degrees of freedom.

Residual Mean Square, MSE, is defined as

$$s^2 = MSE = \frac{SSE}{n - 2}.$$

We use s^2 to estimate σ^2

More Comments about the Simple Linear Regression Model

- The model $Y = \beta_0 + \beta_1 X$ is said to be simple, linear in the parameters, and linear in the predictor variable. Could you tell me whether or not the following models are linear? simple?

- $Y = \beta_0(1 + \beta_1)X + \varepsilon$

- $Y = \beta_0 + \beta_1^X + \varepsilon$

- $Y = \beta_0^2 + \beta_1 X + \varepsilon$

- $Y = \beta_0 + \exp^{\beta_1} X + \varepsilon$

- $Y = \beta_0 + \beta_1 X^3 + \varepsilon$

- $Y = \beta_0 + \beta_1 \exp^X + \varepsilon$

- $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$

Properties of estimators

Recall that our estimators are given by

$$\begin{cases} \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x}; \\ \hat{\beta}_1 &= \frac{S_{xy}}{S_{xx}} = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}. \end{cases}$$

Unbiasedness:

- $$\mathbb{E}[\hat{\beta}_1] = \mathbb{E}\left[\frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{S_{xx}}\right] = \frac{\sum_{i=1}^n (x_i - \bar{x}) \mathbb{E} y_i}{S_{xx}} = \frac{\sum_{i=1}^n (x_i - \bar{x}) (\beta_0 + \beta_1 x_i)}{S_{xx}} = 0 + \frac{\sum_{i=1}^n \beta_1 (x_i - \bar{x})^2}{S_{xx}} = \beta_1.$$
- $$\mathbb{E}(\hat{\beta}_0) = \mathbb{E}(\bar{y} - \hat{\beta}_1 \bar{x}) = (\beta_0 + \beta_1 \bar{x} - \beta_1 \bar{x}) = \beta_0.$$

$\text{Var}(\hat{\beta}_0)$ and $\text{Var}(\hat{\beta}_1)$

- $\text{Var}(\hat{\beta}_1) = \text{Var} \left[\sum_{i=1}^n \frac{(x_i - \bar{x})y_i}{S_{xx}} \right] \text{ (independence) } =$
$$\sum_{i=1}^n \left[\frac{(x_i - \bar{x})^2 \text{Var}(y_i)}{S_{xx}^2} \right] = \left[\frac{(\sum_{i=1}^n (x_i - \bar{x})^2) \sigma^2}{S_{xx}^2} \right] = \frac{\sigma^2}{S_{xx}}$$
- $\text{Var}(\hat{\beta}_0) = \text{Var}(\bar{y} - \hat{\beta}_1 \bar{x}) = \frac{\sigma^2}{n} + \frac{(\bar{x})^2 \sigma^2}{S_{xx}} = \sigma^2 \left(\frac{\sum x_i^2}{n S_{xx}} \right)?$
- $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\sigma^2 \left(\frac{\bar{x}}{S_{xx}} \right)?$

Questions:

- What does it mean $\text{Var}(\hat{\beta}_1) \neq 0$?
- What is the source of randomness in $\hat{\beta}_1$?
- How can you reduce $\text{Var}(\hat{\beta}_1)$?

Goodness of Fit: Evaluating the fit of simple linear regression model

- After we fit the simple regression model, how can we know we fit a good model?
 - 1 Use scatter plot to make sure the relationship between X and Y is linear;
 - 2 Can we have a numerical measure?

ANOVA Decomposition

- Define notations:

$$TSS = \sum_{i=1}^n (y_i - \bar{y})^2, \quad \text{Total sum of squares}$$

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad \text{Residual sum of squares}$$

$$SS_{reg} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2, \quad \text{Sum of squares in regression}$$

where $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$;

- ANOVA Decomposition:

$$\underbrace{TSS}_{\text{Total variations in } Y} = \underbrace{SS_{reg}}_{\text{Variations in } Y \text{ explained by model}} + \underbrace{RSS}_{\text{Variations in } Y \text{ unexplained by model}}$$

Definition of R^2

The R^2 is defined as

$$R^2 = \frac{SS_{reg}}{TSS} = 1 - \frac{RSS}{TSS}$$

- R^2 is between 0 and 1;
- The bigger the R^2 , the better (loosely speaking...);
- For simple linear regression, R^2 equals to r^2 , the sample correlation between X and Y

$$R^2 = r^2 = \frac{\{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})\}^2}{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}.$$

Some comments on R^2

- R^2 can be used to measure the strength of association;
- R^2 can only measure **LINEAR** relationship;
- If the relationship between x and y is not linear, R^2 is useless!
- R^2 and the scatter plot need to be used together!

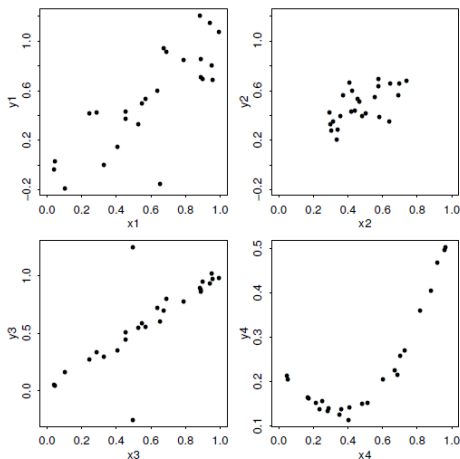


Figure 2.3 Four simulated datasets where R^2 is about 0.65. The plot on the upper

More Comments about the Simple Linear Regression Model

- The three basic assumptions for the simple linear regression model are
 - ε_i is independent. i.e. different subject does not affect each other in error
 - ε_i is identically distributed, i.e. assume the common variance σ^2
 - ε_i follows a normal distribution

In a summary $\varepsilon_i \sim \text{iid } N(0, \sigma^2)$

- With these assumptions, it can be easy to see that the maximum likelihood estimator is exactly the same as least squares estimator (the proof as follows will be skipped).

Inference of β_1

- $\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$ is a function of $Y_1, Y_2, \dots, Y_n$
- $\hat{\beta}_1$ has mean β_1 . i.e. the least square estimate of β_1 is unbiased
- $\hat{\beta}_1$ has variance $\frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}$.
- $\hat{\beta}_1$ is UMVUE, i.e. unbiased and has minimum variance among all other unbiased linear estimator.
- $\hat{\beta}_1 \sim N(\beta_1, \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2})$
- we use $\frac{MSE}{\sum_{i=1}^n (X_i - \bar{X})^2}$ to estimate the variance of $\hat{\beta}_1$
- $\frac{\hat{\beta}_1 - \beta_1}{s\{\hat{\beta}_1\}}$ follows what distribution? t distribution with $n - 2$ df

Hypothesis Testing on β_1

Hypothesis Testing

1. Identify the parameter of interest.
2. State null and alternative hypotheses.

Two-sided test	$H_0 : \beta_1 = \beta_{10}$ versus $H_1 : \beta_1 \neq \beta_{10}$
One-sided test	$H_0 : \beta_1 = \beta_{10}$ versus $H_1 : \beta_1 > \beta_{10}$ or $H_0 : \beta_1 = \beta_{10}$ versus $H_1 : \beta_1 < \beta_{10}$

3. Choose the significant level α . It is usually given, if not, you can choose 0.01 or 0.05.

Hypothesis Testing on β_1

4. Find the test statistic and its sampling distribution under H_0 .

$$T_0 = \frac{\hat{\beta}_1 - \beta_{10}}{s\{\hat{\beta}_1\}}$$

5. Compute $t_0 = \frac{\hat{\beta}_1 - \beta_{10}}{s\{\hat{\beta}_1\}}$.

6. State the rejection region or compute the p-value.

Method1: Use acceptance region or rejection region.

Alternative hyp.	Acceptance region	Rejection region
$H_1 : \beta_1 \neq \beta_{10}$	$(-t_{1-\alpha/2, n-2}, t_{1-\alpha/2, n-2})$	$(-\infty, -t_{1-\alpha/2, n-2})$ or $(t_{1-\alpha/2, n-2}, \infty)$
$H_1 : \beta_1 > \beta_{10}$	$(-\infty, t_{1-\alpha, n-2})$	$(t_{1-\alpha, n-2}, \infty)$
$H_1 : \beta_1 < \beta_{10}$	$(-t_{1-\alpha, n-2}, \infty)$	$(-\infty, -t_{1-\alpha, n-2})$

Hypothesis Testing on β_1

Method2: Use the p-value

Alternative	$H_1 : \beta_1 \neq \beta_{10}$	$H_1 : \beta_1 > \beta_{10}$	$H_1 : \beta < \beta_{10}$
P-value	$2P(T > t_0)$	$P(T > t_0)$	$P(T < t_0)$

7. Decide whether or not the null hypothesis should be rejected.

If the test statistic t_0 is in acceptance region, we cannot reject H_0 .

If the test statistic t_0 is in rejection region, we reject H_0 .

If p-value $> \alpha$, we cannot reject H_0 .

If p-value $< \alpha$, we can reject H_0 .

$(1 - \alpha)$ confidence interval for β_1

$(1 - \alpha)100\%$ confidence interval for β_1 is

Point Estimator $\pm$ (z-value or t-value) $\times$ St Error of Pt Est,

	Confidence interval
$\beta_1 \neq \beta_{10}$	$(\hat{\beta}_1 - t_{1-\alpha/2, n-2} s\{\hat{\beta}_1\}, \hat{\beta}_1 + t_{1-\alpha/2, n-2} s\{\hat{\beta}_1\})$
$\beta_1 < \beta_{10}$	$(-\infty, \hat{\beta}_1 + t_{1-\alpha, n-2} s\{\hat{\beta}_1\})$
$\beta_1 > \beta_{10}$	$(\hat{\beta}_1 - t_{1-\alpha, n-2} s\{\hat{\beta}_1\}, \infty)$

Hypothesis Testing on β_1

- $H_0 : \beta_1 = \beta_{10}$ versus $H_1 : \beta_1 \neq \beta_{10}$

If $\beta_{10} \in (\hat{\beta}_1 - t_{1-\alpha/2, n-2}s\{\hat{\beta}_1\}, \hat{\beta}_1 + t_{1-\alpha/2, n-2}s\{\hat{\beta}_1\})$, then we fail to reject H_0 under significance level α .

- $H_0 : \beta_1 = \beta_{10}$ versus $H_1 : \beta_1 > \beta_{10}$

If $\beta_{10} \in (\hat{\beta}_1 - t_{1-\alpha, n-2}s\{\hat{\beta}_1\}, \infty)$, then we fail to reject H_0 under significance level α .

- $H_0 : \beta_1 = \beta_{10}$ versus $H_1 : \beta_1 < \beta_{10}$

If $\beta_{10} \in (-\infty, \hat{\beta}_1 + t_{1-\alpha, n-2}s\{\hat{\beta}_1\})$, then we fail to reject H_0 under significance level α .

KWH Example

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	27.85107	7.80654	3.57	0.0021
AC	1	5.34108	1.03067	5.18	<.0001

- What is your interpretation for the t test respective to β_1 ?
- What hypothesis test does this t test to test for?

$$H_0 : \beta_1 = 0 \text{ versus } H_1 : \beta_1 \neq 0$$

Since p-value less than 0.05, reject H_0 at the level of 0.05. i.e. there is strong evidence to say that there exists a linear relation between KWH and AC.

KWH Example

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	27.85107	7.80654	3.57	0.0021
AC	1	5.34108	1.03067	5.18	<.0001

- If I don't give you the standard deviation of $\hat{\beta}_1$ and the t value, are you able to compute them?

We use $\frac{MSE}{\sum_{i=1}^n (X_i - \bar{X})^2}$ to estimate the variance of $\hat{\beta}_1$. So, standard error of $\hat{\beta}_1 = \sqrt{\frac{208.88994}{20 * 3.13562^2}} = 1.03$

$$t_0 = \frac{\hat{\beta}_1 - 0}{s(\hat{\beta}_1)} = \frac{5.34}{1.03} = 5.18$$

Hypothesis Testing on β_1

- What is the 95% confidence interval for β_1 ? How to interpret?

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$$5.34 \pm t_{.975, 19}(1.03) = 5.34 \pm 2.093(1.03) = (3.18, 7.496)$$

With confidence coefficient 0.95, we estimate that the mean electricity consumption is between 3.16 and 7.496 KWH for each additional hour of AC used

- What is your conclusion on the test based on the confidence interval?

Hypothesis Testing on β_1

- What is the 95% confidence interval for β_1 ? How to interpret?

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With confidence coefficient 0.95, we estimate that the mean electricity consumption is between 3.16 and 7.496 KWH for each additional hour of AC used

- What is your conclusion on the test based on the confidence interval?

Since 0 is not contained in the confidence interval (3.18, 7.496), reject H_0 at the level of 0.05. i.e. there is strong evidence to say that there exists a linear relation between KWH and AC.

- Do you think that the slope is greater than 3? How about the slope greater than 4?

The %95 confidence interval for β_1 for one side test is

$$(\hat{\beta}_1 - t_{1-\alpha, n-2} s\{\hat{\beta}_1\}, \infty) = (5.34 - t_{0.95, 19} \cdot 1.03, \infty)$$

$= (5.34 - 1.729 \cdot 1.03, \infty) = (3.56, \infty)$. Since 3 is not included in the CI, and 4 is included in the CI, we reject $H_0 : \beta_1 = 3$ but fail to reject $H_0 : \beta_1 = 4$.

Hypothesis Testing on β_1

- For the special case: $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$, you can use another method
- Interpretation: under the null hypothesis $\beta_1 = 0$, it implies not only there is no linear association between Y and X , but also that there is NO RELATION between Y and X
- Test Statistics is $F_0 = \frac{MSR}{MSE} \sim F_{1,n-2}$ under H_0
- If $F_0 > F_{1-\alpha,1,n-2}$ or $p\text{-value} = p(F > F_0) < \alpha$, then we reject the null hypothesis.
- F test only works for $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$. i.e. only under this special test, F and t are both applicable
- t test can work for all other tests. For example, for any value β_{10} may not be zero. Or right tail test ($>$) and left tail test ($<$).
- For the special test $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$, if α is fixed, then $F_0 = t_0^2$

KWH Example

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	5609.66260	5609.66260	26.85	<.0001
Error	19	3968.90883	208.88994		
Corrected Total	20	9578.57143			

Simple Statistics						
Variable	N	Mean	Std Dev	Sum	Minimum	Maximum
AC	21	6.92857	3.13562	145.50000	1.50000	13.50000
KWH	21	64.85714	21.88444	1362	17.00000	94.00000

- What is your interpretation for the F test in the table?
- What hypothesis test does this F test to test for? $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$

Since p-value less than 0.05, reject H_0 at the level of 0.05. i.e. there is strong evidence to say that there exists a linear relation between KWH and AC.

Inference of $E(Y_h)$

- Let X_h denotes some specific value of the predictor. X_h may be a value that already observed in the data, or it may be other value within the scope of the model that has not occurred yet
- $\hat{Y}_h = \hat{\beta}_0 + \hat{\beta}_1 X_h = \bar{Y} + \hat{\beta}_1 (X_h - \bar{X})$ is the point estimator of $E(Y_h)$
- $\hat{Y}_h$ has mean $E(Y_h) = \beta_0 + \beta_1 X_h$
- $\hat{Y}_h$ has variance $[\frac{1}{n} + \frac{(X_h - \bar{X})^2}{\sum (X_i - \bar{X})^2}] \sigma^2$. The variability of $\hat{Y}_h$ is affected by how far X_h is from $\bar{X}$. The further from $\bar{X}$ is X_h , the larger the variance of $\hat{Y}_h$. The variance is smallest when $X_h = \bar{X}$
- $\hat{Y}_h \sim N(\beta_0 + \beta_1 X_h, [\frac{1}{n} + \frac{(X_h - \bar{X})^2}{\sum (X_i - \bar{X})^2}] \sigma^2)$
- we use $[\frac{1}{n} + \frac{(X_h - \bar{X})^2}{\sum (X_i - \bar{X})^2}] MSE$ to estimate the variance of $\hat{Y}_h$
- $\frac{\hat{Y}_h - E(Y_h)}{s\{\hat{Y}_h\}}$ follows t distribution with $n - 2$ df

Prediction interval for a new trial

- Let X_h denotes a value of a new trial within the scope of the model that has not happened yet
- Let $Y_{h(new)}$ denotes the new observation of Y when $X = X_h$
- We assume that the same underlying regression model continues to be appropriate for the new observation
- We can only predict $Y_{h(new)}$ by $\hat{Y}_h$
- The variance of $Y_{h(new)}$ contains: 1) variance of $\hat{Y}_h$, i.e. possible distribution of the fitted response value; 2) variance of the distribution of observation. So, $\sigma^2(pred) = \sigma^2(Y_{h(new)}) + \sigma^2(\hat{Y}_h)$
- We use $s^2(pred) = [1 + \frac{1}{n} + \frac{(X_h - \bar{X})^2}{\sum (X_i - \bar{X})^2}] MSE$ to estimate $\sigma^2(pred)$
- $\frac{Y_{h(new)} - \hat{Y}_h}{s(pred)}$ follows t distribution with $n - 2$ df
- The $(1 - \alpha)100\%$ prediction interval of $Y_{h(new)}$ is (only hold for one single prediction)

$$(\hat{Y}_h - t_{1-\alpha/2, n-2} s(pred), \hat{Y}_h + t_{1-\alpha/2, n-2} s(pred))$$

- Prediction interval is wider than the Confidence Interval for the same predictor value

More Comments about the Simple Linear Regression

- If Y only depart from normality slightly, $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{Y}_h$ will be approximately normal, and t test can still applied to get decision and confidence interval. If the response data are substantially far from normal but with large sample size, $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{Y}_h$ will by asymptotic normal and the same decision rule and CI are still applicable if replace t by z.
- We assume that the predictor X is constant. But if X is also random variable, then the estimates are generally no longer unbiased
- The prediction interval for a new observation, unlike the CI for the mean response, are sensitive to departures from normality.
- The formulas that we used here to compute prediction interval only works for one single point or several points with the same value of X_h
- If the new X_h is far away from the range of past data, the regression function that fits the past data might not appropriate for prediction

Multiple Linear Regression

Gauss-Markov Theorem

Or, more generally, for an arbitrary vector $\mathbf{t}$

Theorem 2

(Gauss-Markov Theorem) Suppose that $\boldsymbol{\mu} = E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Var}(\mathbf{Y}) = \sigma^2\mathbf{I}$. Let $\tilde{\phi} = \mathbf{c}^T\mathbf{Y}$ be any unbiased linear estimator of $\phi = \mathbf{t}^T\boldsymbol{\beta}$, where $\mathbf{t}$ is an arbitrary vector. Then

$$\text{var}(\tilde{\phi}) \geq \text{var}(\hat{\phi})$$

where $\hat{\phi} = \mathbf{t}^T\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}$ is the least squares estimator of $\boldsymbol{\beta}$. Notice that, since $\mathbf{t}$ is arbitrary, this theorem implies that each element of $\hat{\boldsymbol{\beta}}$ is a minimum variance unbiased estimator.

When should we avoid using least squares estimators

Situations where estimators other than OLS should be considered are:

- When assumptions are not satisfied, such as linearity
- When the errors are correlated or have unequal variance
- When the error distribution is long-tailed
- When the predictors are highly correlated

Comment on identifiability

When $\mathbf{X}$ is not of full column rank, $\mathbf{X}^T \mathbf{X}$ will be singular and cannot be inverted. There will be infinitely many solutions to β and β is at least partially unidentifiable.

$\mathbf{X}$ is not of full column rank when its columns are linearly dependent.

Some examples for the same could be measuring the same quality in two different units, having columns which can be mathematically expressed as a linear combination of other columns, or when $p > n$.

Statistics packages handle non-identifiability differently. R for example, will give you a warning or say NA in place of coefficient estimates. If you get these errors or warning, it is usually advisable to understand the reason behind such a warning and to take rectifying measures. (We will handle the $p > n$ case later in the course).

More problematic are cases where we are close to unidentifiability. But some corrective measures could be to have a look if the signs of your coefficient makes sense, if your standard errors do not look abnormally high, etc.

Comment on orthogonality

We typically like $\mathbf{X}$ to be such that $\mathbf{X}^T \mathbf{X}$ is a diagonal matrix. But this is hardly in control of experimenter, especially in observational studies.

In a designed experiment, however, we choose $\mathbf{X}$ so that the orthogonality property is satisfied. Suppose we partition our $\mathbf{X}$ into two partitions, then with orthogonality, we can identify the coefficients of the variables in the first component regardless of whether the other set of variables are in the model or not.

The above decouples the estimates but this decoupling cannot be extended to testing or related inference. Nevertheless it is a reasonable property to help designed experiments.

A major problem with R^2

Recall the definition of R^2

$$R^2 = 1 - \frac{RSS}{TSS}$$

$$RSS(\beta_0, \dots, \beta_p) = \min_{\beta} \sum_{i=1}^n (y_i - \beta_0 - x_{1i}\beta_1 - \dots - x_{pi}\beta_p)^2.$$

$$TSS = \sum_{i=1}^n (y_i - \bar{y})^2.$$

Suppose we add an addition predictor x_{p+1} into our regression model (which can have nothing to do with the response variable Y at all), then the new fitted RSS would be

$$RSS(\beta_0, \dots, \beta_p, \beta_{p+1}) = \min_{\beta} \sum_{i=1}^n (y_i - \beta_0 - x_{1i}\beta_1 - \dots - x_{pi}\beta_p - x_{p+1,i}\beta_{p+1})$$

- $RSS(\beta_0, \dots, \beta_p, \beta_{p+1}) \leq RSS(\beta_0, \dots, \beta_p)$ is always true;
- TSS does not change when adding a new predictor;
- R^2 will always increase by adding a new predictor (no matter what predictor!);
- R^2 cannot detect overfitting!

Adjusted R^2

Because of the defect of R^2 , it may not be a good measure for evaluating the goodness of fit

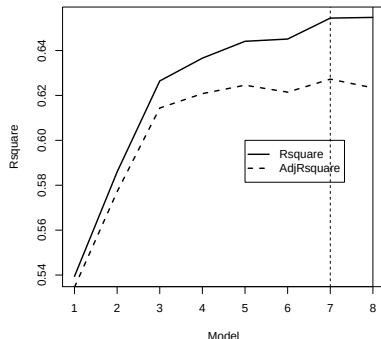
- Define the adjusted R^2 AS

$$R_{adj}^2 = 1 - \frac{RSS/(n - p - 1)}{TSS/(n - 1)} = 1 - \frac{MSE}{MST},$$

- $df_1 = n - p - 1$ is the degrees of freedom of RSS (the regression model with $p + 1$ parameters);
- $df_2 = n - 1$ is the degrees of freedom of TSS (the null model with only an intercept!)
- MSE: stands for mean square error;
- MST: stands for mean square total error;
- R_{adj}^2 For a “bigger” model (with a larger p) will be penalized;
- R_{adj}^2 : maybe less than 0; (when would that happen?)

Model complexity V.S R^2

- 1 The model should fit your data well;
- 2 The model should be simple; (better interpretation.)



- There seems to be a trade off between “model simplicity” (# of predictors) and “model fit” (R^2);
- We are “buying” R^2 using “model simplicity” as our currency;
- We want a good deal!

Model complexity V.S R^2

- What is a good deal then?
- Sacrifices less “model simplicity” for more “goodness of fit” R^2 ;
- For each added predictor, we want a “big” improvement in R^2 ;
- Recall the definition of R^2

$$R^2 = 1 - \frac{RSS}{TSS}.$$

- For each added predictor, we want a “big” reduction in RSS ;
- But how big is big enough?
- Consider the measure

$$\frac{RSS_{small} - RSS_{big}}{RSS_{big}}$$

- If $RSS_{small} - RSS_{big} > 0$ is small, then the fit of the smaller model is almost as good as the bigger model, then **pick the smaller model on the grounds of simplicity**

Comparison rules

- The comparison rules should be something like **pick the larger model if**

$$\frac{RSS_{small} - RSS_{big}}{RSS_{big}} > c$$

for some $c > 0$.

- Do a little re-scaling, define

$$F = \frac{(RSS_{small} - RSS_{big}) / (df_{small} - df_{big})}{RSS_{big} / df_{big}},$$

where

- df_{small} : the degrees of freedom of the **RSS** in smaller model (n-# of parameters in the “small model”);
 - df_{big} : the degrees of freedom of **RSS** in the bigger model (n-# of parameters in the “bigger model”);
 - $df_{big} - df_{small}$: difference in number of parameters between “small” and “bigger” model.
- Interpretation of the numerator: average RSS reduction per additional parameter—**Mean Square**.

The F-distribution

Under several assumptions

- 1 The “small” model must be part of the “bigger” model;
 - The professional term: the small model is **nested** in the bigger model;
- 2 The null hypothesis (roughly speaking): the “small” model is truly as good as the “bigger” model

Then,

$$F = \frac{(RSS_{small} - RSS_{big}) / (df_{small} - df_{big})}{RSS_{big} / df_{big}} \sim F_{df_{small} - df_{big}, df_{big}}$$

- A very large value of F will invalidate our **null hypothesis: the “small” model is truly as good as the “bigger” model**;
- A very large value of F indicate (with a big chance), the bigger model is better;
- Critical value $F_{df_{small} - df_{big}, df_{big}}$ can be found in a F-table or the R function “qf()”.

General Linear Test Approach

This approach works for two nested model, i.e. the model of H_0 (named reduced model) is nested under the model of H_1 (named full model). The steps for this approach are

- 1) From the statement of the hypothesis, write down the corresponding model and judge whether the model under H_0 is nested under the model under H_1 . If nested, continue. Or else use other approach
- 2) Fit the Full model and compute $SSE(F)$
- 3) Fit the Reduced model, and compute $SSE(R)$
- 4) Compute the test statistic

$$F_0 = \frac{SSE(R) - SSE(F)}{df_R - df_F} \div \frac{SSE(F)}{df_F}$$

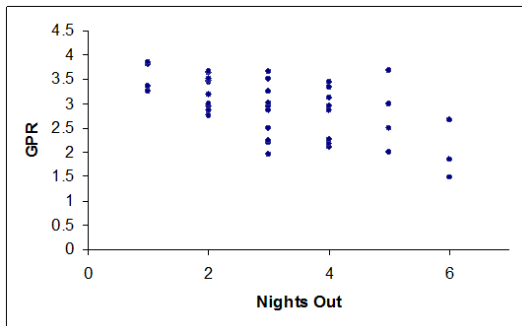
- 5) If $F_0 > F_{1-\alpha, df_R - df_F, df_F}$, then reject H_0 at the level of α

What the F-test cannot do?

- F-test cannot compare two models that are **not nested**.
- You full model must be good enough to use the F-test (i.e. the full model must contain all significant predictors;)
- F-test cannot test $H_0 : \beta_j \beta_k = 1$; (only linear combinations of β_k 's can be tested using F-test)
- F-test also cannot handle the one-sided alternative statement

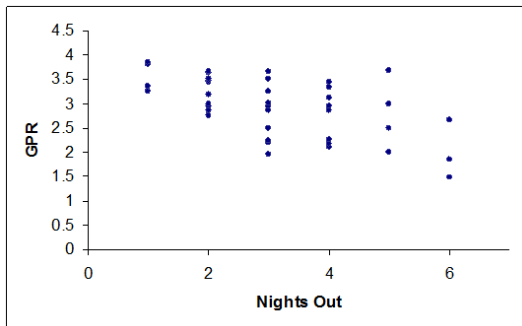
Detecting linear association: scatter plot

- Test the claim that the number of times a BU student goes out per week is negatively related to their GPA.
- Explanatory variable: Nights out per week;
- Response variable: GPA
- Does there appear to be an association? If so, describe it.



Detecting linear association: scatter plot

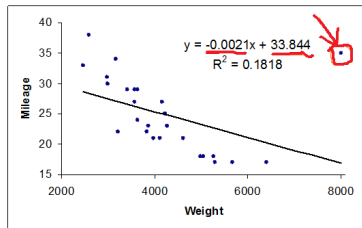
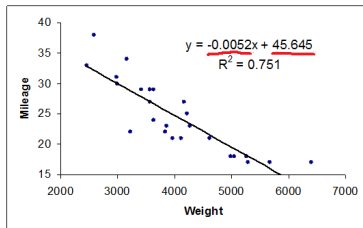
- Test the claim that the number of times a BU student goes out per week is negatively related to their GPA.
- Explanatory variable: Nights out per week;
- Response variable: GPA
- Does there appear to be an association? If so, describe it.



- There appears to be a weak negative association.

Detecting influential point/outlier: scatter plot

- **An outlier** is an observation that does not follow the general pattern of the rest of the data;
 - It does not fall in the straight-line pattern with the rest of the data.
- Outliers can have a strong impact on the least square **estimators** of slope and intercept of the regression line.
- If your data has a potentially influential point, you should
 - Investigate the point. Is it possible that it was recorded incorrectly?
 - Refit the regression line without the point to see if it has a large effect.

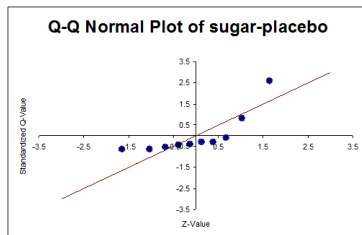
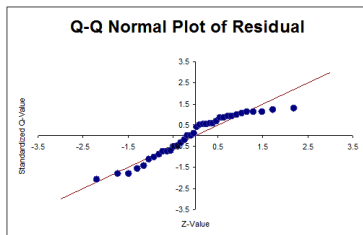


Check for normality of the residuals: Q-Q plot

- After the regression line is fitted, calculate the residuals by

$$\hat{\varepsilon}_i = y_i - \hat{y}_i$$

- Plot the Q-Q plot of the residuals $\hat{\varepsilon}$'s;
 - You can use R function `qqnorm()` to draw this plot;
- If at each x value, the distribution of the y 's is normal, this plot should be **a straight line**. (Left pic: normal; right pic: non-normal)

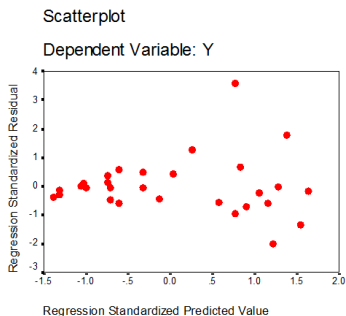
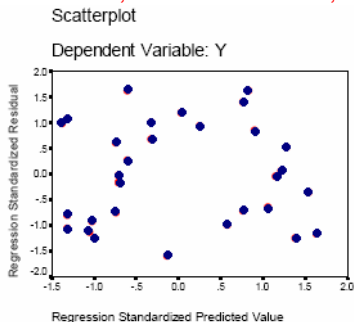


Check for constant variance of residuals: the residual plot

- After the regression line is fitted, calculate the residuals by

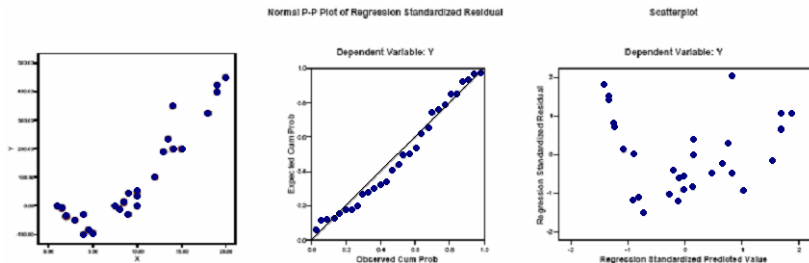
$$\hat{\varepsilon}_i = y_i - \hat{y}_i$$

- Plot the residual plot of the residuals $\hat{\varepsilon}_i$'s: the predicted value $\hat{y}_i$ as the x -axis and the residual e_i as the y -axis;
- But why do we choose to use the predicted value $\hat{y}_i$ as the x -axis? (recall: $\hat{\varepsilon}$ is orthogonal to $\hat{Y}$ under regression model assumptions.)
- If the constant variance assumption is met, the residual plot should look random; Left: look random; Right: there is a fan shape pattern:



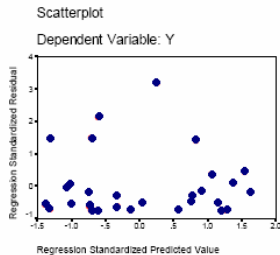
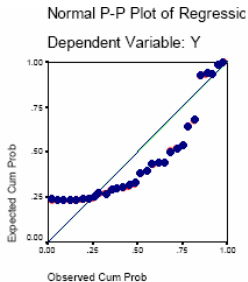
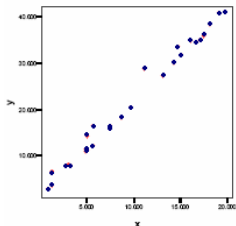
Potential problem # 1

- The relationship between x and y is not linear.



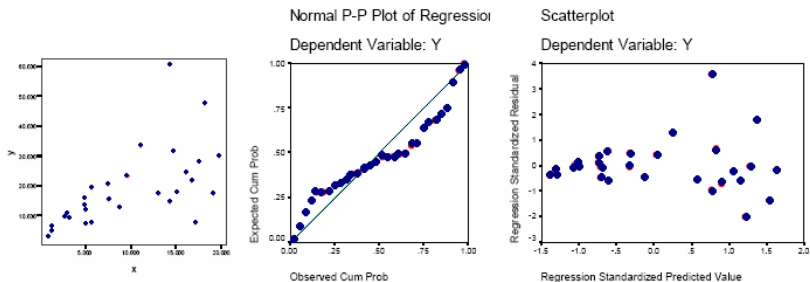
Potential problem # 2

- The distribution of the response variable (y) is not normal for each value of the explanatory variable (x).



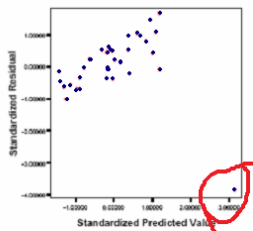
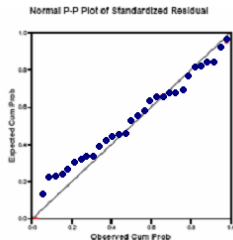
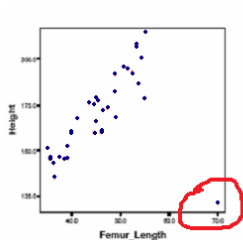
Potential problem # 3

- The variance/standard deviation is not the same for each x value.



Potential problem # 4

- There is an influential point that is obscuring the relationship between x and y .



Assumptions on the error term ε_i

- For the linear regression model

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi} + \varepsilon_i, \quad i = 1, \dots, n$$

- Conditions on ε_i :

- 1 Constant variance: $\text{Var } \varepsilon_i = \sigma^2$ for all $i = 1, \dots, n$;
- 2 ε_i are normally distributed (only assumed for confidence intervals and hypothesis tests)
- 3 All error terms ε_i 's are independent;

- We have learned how to detect the problems, but then how to fix them?

Detection tools for non-constant variance

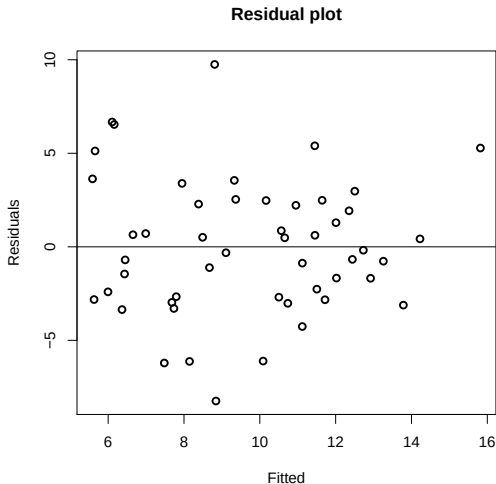
- The first tool for detecting the non-constant variance is the residual plot

$\hat{\varepsilon}_i$ (residuals) v.s (the fitted values) $\hat{y}_i$

- If no problem, you should observe a **random** pattern;
- R code for generating residual plots;

```
data(savings,package="faraway")
lmod <- lm(sr~pop15+pop75+dpi+ddpi,savings)
res<-residuals(lmod)#extract residuals
fit<-fitted(lmod)#extract fitted values
#the residual plot
plot(fit,res,xlab="Fitted",ylab="Residuals",main="Residual plot",lwd=2)
abline(h=0) #draw the reference line
```

Residual plots for “savings” data

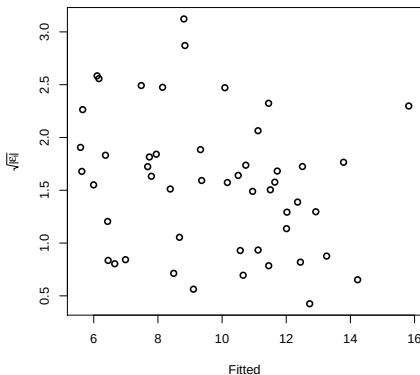


- We do not observe anything suspicious!
- Just to double check, we can use the plot using $\sqrt{|\hat{\epsilon}_i|}$ v.s. $\hat{y}_i$ (more accurate);

Detecting non-constant variance: plot $\sqrt{|\hat{\epsilon}_i|}$ v.s. $\hat{y}_i$

R codes:

```
plot(fit,sqrt(abs(res)),xlab="Fitted",  
ylab=expression(sqrt(abs(epsilon[i]))),lwd=2)  
abline(h=0) #draw the reference line
```



- Again, we do not observe anything suspicious!
- The constant variance assumption is probably ok for this model.

Why would there be non-constant variance?

- One of the most important reason for non-constant variance: **The Ey and $Var(y)$ have certain relationships!**
- Recall that the regression model $Y = X\beta + \varepsilon$ we have $EY = X\beta$ and $Var(Y) = \sigma^2 I$. So Ey and $Var(y)$ should have no relationship at all;
- However, in practice, some times ε may be correlated with X . In econometrics, this is called **Endogeneity**;
- Assume that the “true” model is $y_i = \alpha + \beta x_i + \gamma z_i + u_i$;
- But we omit z_i (perhaps because we don't have a measure for it) when we run our regression;
- z_i will get **absorbed** by the error term and we will actually estimate

$$y_i = \alpha + \beta x_i + \varepsilon_i, \text{ where } \varepsilon_i = \gamma z_i + u_i;$$

- If the correlation of x and z is not 0 and z separately affects y (meaning $\gamma \neq 0$), then x is correlated with the error term ε .
- Another simple example: if y is poisson, then $E(y) = Var(y) = \lambda$! Mean equals variance $\Rightarrow$ variance changes as the mean change!

Remedies for non-constant variance of ε

There are couple of remedies for non-constant variance of ε

- 1 Weighted least square estimator: we will talk about this later;
- 2 Transformation of y :
 - This maybe the most sensible thing to do in practice;
 - Commonly used transformations:

$$\sqrt{y}, \quad \log(y), \quad \log(y + c), \quad 1/y$$

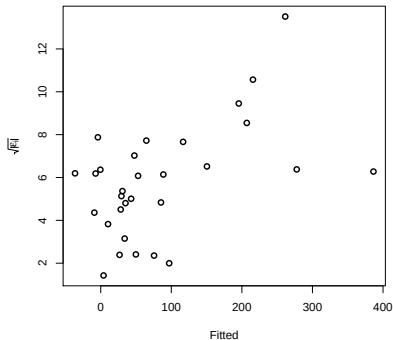
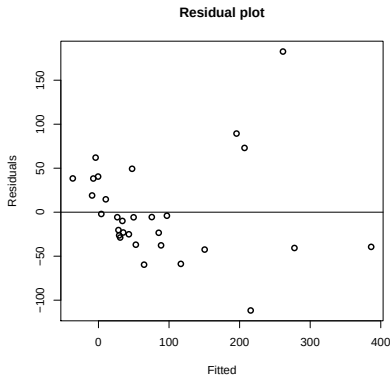
- How to decide which one to use? Try!
- Try all commonly used transformations and see which one solves your problem!

Residual plots for “gala” data

R codes

```
data(gala,package="faraway")
mod<-lm(Species~Area+Elevation+Scruz+Nearest+Adjacent, gala)
res<-residuals(mod)#extract residuals
fit<-fitted(mod)#extract fitted values
#the residual plot
plot(fit,res,xlab="Fitted",ylab="Residuals",main="Residual plot",lwd=2)
abline(h=0) #draw the reference line
##square-root residual plot
plot(fit,sqrt(abs(res)),xlab="Fitted",
ylab=expression(sqrt(abs(epsilon[i]))),lwd=2)
abline(h=0) #draw the reference line
```

Residual plots for “gala” data



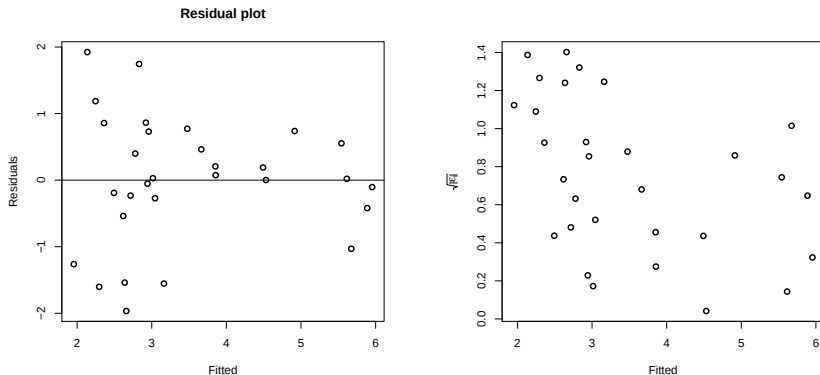
- Both of them show some sort of associations!
- Try transformations!

Residual plots after a log transformation

R codes

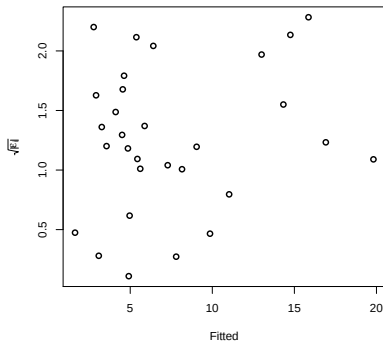
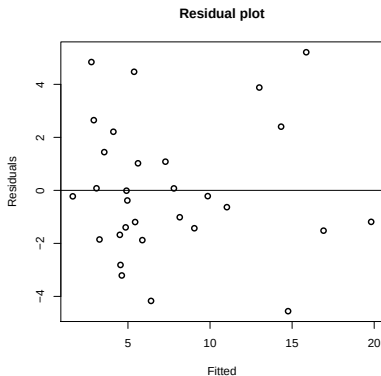
```
data(gala,package="faraway")
lmod<-lm(log(Species)~Area+Elevation+Scruz+Nearest+Adjacent, gala)
res<-residuals(lmod)#extract residuals
fit<-fitted(lmod)#extract fitted values
#the residual plot
plot(fit,res,xlab="Fitted",ylab="Residuals",main="Residual plot",lwd=2)
abline(h=0) #draw the reference line
##square-root residual plot
plot(fit,sqrt(abs(res)),xlab="Fitted",
ylab=expression(sqrt(abs(epsilon[i]))),lwd=2)
```

Residual plots after a log transformation



- It is somewhat better than before, but still not quite right!
- Try another transformation: $\sqrt{y}$

Residual plots after a square-root transformation



- Looks almost random!
- $\sqrt{y}$ seems ok
- But you may not get lucky every time.
- But even you cannot completely remove the problem, it does not hurt to make the situation less severe by using some transformation.

The second problem with ε : the normality

- The normality is only required for confidence intervals and hypothesis testing;
 - Recall the derivation of the t-test and F-test, we used the normal assumption;
 - Recall that $\hat{\beta}_{ols} = \beta + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \varepsilon$;
 - If the sample size is very large, using central limit theorem will give similar distributions for our test statistics even if the distribution of ε is not normal;
 - However, when you only have small number of observations, $\hat{\beta}_{ols}$ may still be far from normal!
 - When we do statistical analysis, inference is a big part of the process; on the other hand, your sample size is only limited!
 - Therefore, we need normality after all;
- How to check for normality? **Check the q-q plot for residuals $\hat{\varepsilon}_i$'s.**
 - If the qq-plot displays a straight line, then the normality is OK.
 - Otherwise, we need to do something about it;

Normality test: Shapiro-Wilk test

- Shapiro-Wilk test can be used to test the “normality” of the “residuals”;
- The R-command is: `shapiro.test()`;
- The null hypothesis: the data is normally distributed;
- A small p-value means rejection of the normality;
- **Shapiro-Wilk is often too sensitive!** If your data size n is large, it would reject very small departure from normality; (if you accumulate enough data, you can always reject the null hypothesis.)
- **Recommendations:** We can only recommend this in conjunction with a Q-Q plot at best;
- Q-Q plot is still the main thing you should rely on: remember,
 - Small departure from normal is often OK since we have central limit theorem backing us up;
 - Significant departure from normality is the thing we have to worry about;

QQ plot for “savings” data

The R code for generating qq plots;

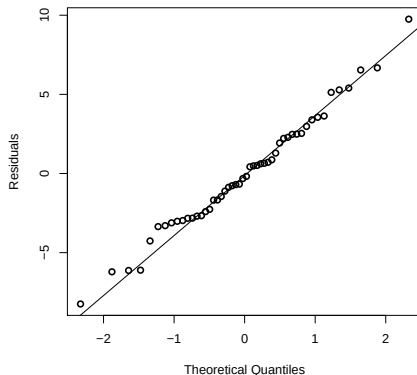
```
data(savings,package="faraway")
lmod <- lm(sr~pop15+pop75+dpi+ddpi,savings)
res<-residuals(lmod)#extract residuals
#the qq plot
qqnorm(res,ylab="Residuals",main="",lwd=2)
qqline(res) #draw a reference line
```

```
##Do Shapiro-Wilk test
shapiro.test(res)
> shapiro.test(res)
```

Shapiro-Wilk normality test

```
data:  res
W = 0.98698, p-value = 0.8524
```

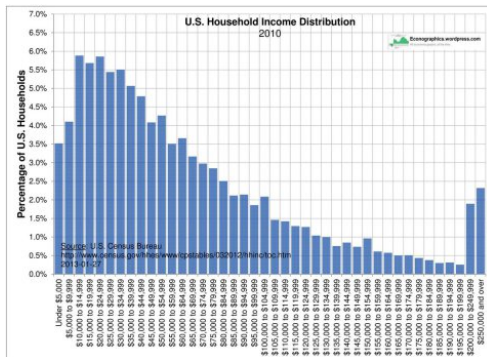
QQ plot for “savings” data



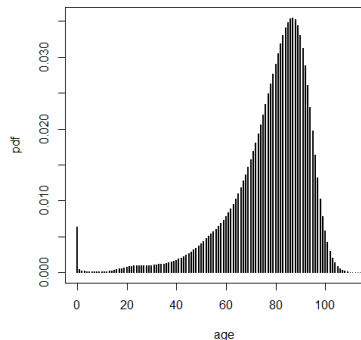
- The QQ-plot is pretty much a straight line;
- Normality seems to be ok;
- In addition, the Shapiro-Wilk test gives a $p\text{-value}=0.8524$, which indicates normality is confirmed;

Some non-normal distributions: skewed distributions

- Skewed distributions: right-skewed (family income): less likely to have large values than small values; tail towards right;
- left-skewed distribution (life-expectancy): less likely to have small values than large values; tail towards left;

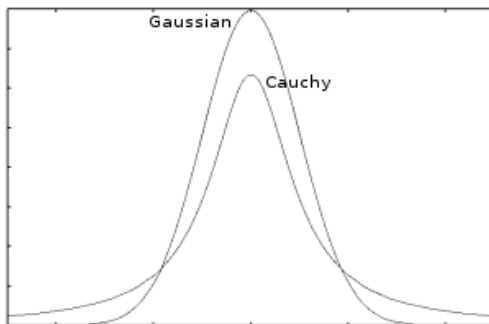


U.S. life expectancy, 2009



Some non-normal distributions: symmetric heavy-tailed distributions

- More likely to have extremely small or large outliers;
- Examples include: Student-t distribution, Cauchy distribution;
- Real examples: innovation process for stock prices;



Some non-normal distributions: symmetric short-tailed distributions

- Has a lighter tail than the normal distribution;
- Less commonly seen;
- See following qq-plots

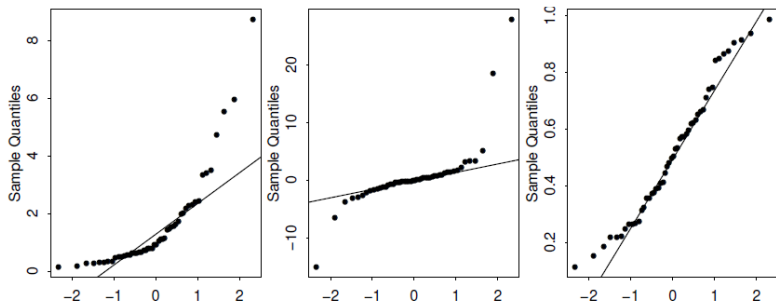


Figure 6.6 Q-Q plots of simulated data for, respectively, a skewed distribution, a long-tailed distribution and a short-tailed distribution.

Recommendations when ε is not normally distributed

The recommendations are case by case

- For a skewed distribution: we can apply a transformation to the response y to make it more normal;
 - One of the most useful tool is the box-cox transformation;

$$y_i^{(\lambda)} = \begin{cases} \frac{y_i^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0, \\ \ln(y_i) & \text{if } \lambda = 0, \end{cases}$$

- The parameter λ needs to be picked by other methods;
- For a **heavy-tailed distribution**:
 - Try to remove some outliers to see if that solves the problem;
 - More robust estimator than the least square estimator should be used;
 - For example, we can use Least Absolute Deviations (LAD), Huber's loss function, or the quantile regression estimator,etc.
- For **short-tailed distribution**: do nothing. Won't have big impact;

A third problem: ε_i 's are not independent

- This can only be checked when your data is naturally ordered in time or space:
- Time series data:
- Spatial data:
- You only need to worry about this issue if your response variable have such features;

How to define influential points/outliers?

- If there is only one predictor in the regression model, identifying the influential points are straightforward;
 - By looking at a scatter plot of y v.s. x ;
- But when we have multiple predictors, how to define influential points/outliers?
- The textbook uses a concept of **leverage**;
- For the i th observation, the (i, i) th element in the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$, denoted as $h_i = H_{ii}$, is defined as its leverage;
- Each observations has its own leverage, $h_1, h_2, \dots, h_n$;
- We can show that $h_1 + h_2 + \dots + h_n = \text{trace}(\mathbf{H}) = p$;
- **A rough rule is that observations with a leverage of more than $2p/n$ should be looked at more closely.** That is, they are influential points; but why?

What are leverages?

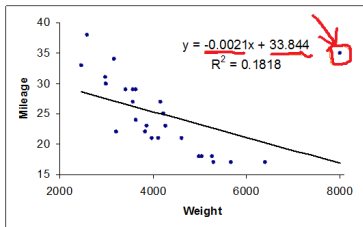
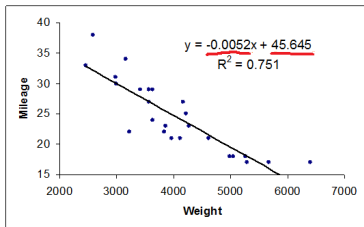
- The leverages h_i 's are diagonal elements of $\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$;
- Recall that $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$, consider the i th observation, we have

$$\hat{y}_i = H_{i1}y_1 + H_{i2}y_2 + \cdots + H_{ii}y_i + \cdots + H_{in}y_n$$

- Hence the leverage $h_i = H_{ii}$ is the amount of weight of the predicted value y_i put on $\hat{y}_i$;
- The leverage h_i is a measure of **self-influence** of y_i on $\hat{y}_i$;
- Note also that the covariance between $\hat{y}_i$ and y_i is

$$\text{cov}(\hat{y}_i, y_i) = H_{ii}\sigma^2 = h_i\sigma^2.$$

- The predicted value $\hat{y}_i$ of an outlier/influential point is heavily impacted by itself; (See the picture)
- Hence the outlier/influential point tend to have big leverage h_i !

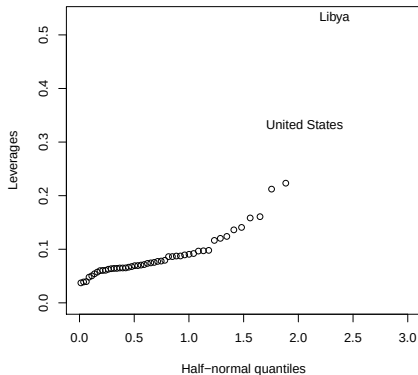


Graphical representation of the leverages

- We have mentioned that **observations with a leverage of more than $2p/n$ should be looked at more closely.**
- The R function for extracting leverages: *hatvalues()*;
- We can also use a graphical tool to pick out abnormal leverages: **Half-normal plot**;
- Half-normal plot is similar to a Q-Q plot, except that it uses quantiles of half-normal distribution as x-axis;
- The R function for generating **leverage-half-normal plot**: *halfnorm()* in package Faraway;

```
library(faraway)
data(savings,package="faraway")
lmod <- lm(sr ~ pop15 + pop75 + dpi + ddpi, savings)
lev <- hatvalues(lmod) #extract the leverage values
countries <- row.names(savings)
halfnorm(lev, labs=countries, ylab="Leverages", xlim=c(0,3))
```

Graphical representation of the leverages



- We are not looking for a linear pattern in the half-normal plot;
- Identify outliers/influential points by picking out abnormal points in this plot!
- In this case, **Libya and United States** seems to be very influential/abnormal;

Another use of leverages: standardized residuals

- Recall that the residual vector $\hat{\epsilon} = (\mathbf{I} - \mathbf{H})\mathbf{Y}$;
- The distribution $\hat{\epsilon} \sim MVN(\mathbf{0}, \sigma^2(\mathbf{I} - \mathbf{H}))$;
- This means the variance of each residual $\hat{\epsilon}_i$ is

$$Var(\hat{\epsilon}_i) = \sigma^2(1 - H_{ii}) = \sigma^2(1 - h_i)$$

- h_i 's are different in general, hence $\hat{\epsilon}_i$'s do not have a **common variance**;
- Recall that when drawing Q-Q plot, we are actually checking whether $\hat{\epsilon}_i$'s has a normal distribution, which would be problematic if $\hat{\epsilon}_i$'s even do not have a common variance;
- A remedy is to define a **standardized residuals** as

$$r_i = \frac{\hat{\epsilon}_i}{s\sqrt{1 - h_i}}$$

where $s = \sqrt{\frac{1}{n-p-1}\hat{\epsilon}^T\hat{\epsilon}}$;

- We have $Var(r_i) \approx 1$ and $corr(r_i, r_j)$ tend to be small;
- In other words, we have a common variance and smaller correlation (closer to independence);

QQ-plot for regular and standardized residuals

```
library(faraway)
data(savings,package="faraway")
lmod <- lm(sr ~ pop15 + pop75 + dpi + ddpi, savings)
res <- residuals(lmod) #regular residuals
stdres <- rstandard(lmod)#standardized residuals
qqnorm(res,pch=16) #qq-plot for regular residuals
qqline(res)
qqnorm(stdres,pch=16) #qq-plot for standardized residuals
qqline(stdres)
```

QQ-plot for regular and standardized residuals

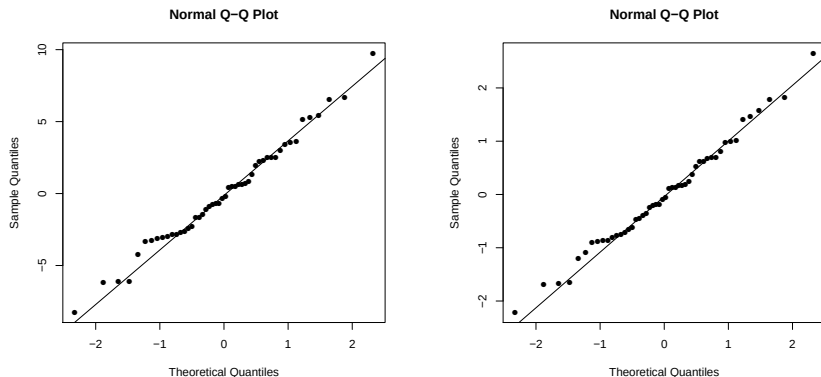


Figure: Left: qq-plot of regular residuals; Right: qq-plot of standardized residuals

- Usually there won't be a big difference;
- QQ plot for standardized residuals is slightly better;
- But it is always good to keep standardized residuals in your mind;

A test for outliers

FINDING UNUSUAL OBSERVATIONS

87

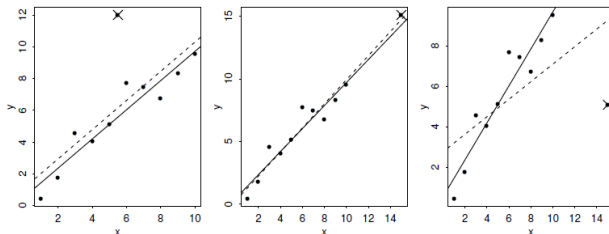


Figure 6.9 In all three plots, the additional point is marked with a cross. The solid line is the fit using the 10 original points and the dashed line is the fit with the added point.

- Let $\hat{\beta}_{(i)}$ and $s_{(i)}$ be the ols estimator and the estimated error standard deviation of the regression model with i th case **deleted**;
- Then the prediction of y_i with the i th case deleted is $\hat{y}_{(i)} = x_i^T \hat{\beta}_{(i)}$;
- For outliers, $\hat{y}_{(i)} - y_i$ is large; (You need be careful here: the underlying assumption is after removing the outlier, all remaining data follow the same model, **which might not be true if you have several outliers; especially when they are close to each other**)

A test for outliers

- So to test whether a point $(\mathbf{x}_i, y_i)$ is an outlier, essentially we need a measure of how large $\hat{y}_{(i)} - y_i$ is;
- Define the statistic (named **studentized residual**)

$$t_i = \frac{\hat{y}_{(i)} - y_i}{se(\hat{y}_{(i)} - y_i)} = \frac{\hat{y}_{(i)} - y_i}{s_{(i)} \sqrt{1 + \mathbf{x}_i^T (\mathbf{X}_{(i)}^T \mathbf{X}_{(i)})^{-1} \mathbf{x}_i}} \sim t_{df=n-p-2}$$

note $df = n - p - 2$ is different from the book!

- Calculation of t_i 's are not easy, we need to estimate the regression n times; (each time leave on observation/case out);
- Luckily, we have a short-cut formula for t_i

$$t_i = \frac{\hat{\varepsilon}_i}{s_{(i)} \sqrt{1 - h_i}} = r_i \sqrt{\frac{n - p - 2}{n - p - 1 - r_i^2}}$$

where $r_i = \frac{\hat{\varepsilon}_i}{s \sqrt{1 - h_i}}$ is the **standardized residual** (2nd appearance!)

Use the studentized residual to find outliers

- We can calculate a t_i for each data points;
- R function for that: `rstudent()`;
- We said a large value of t_i indicate an outlier, but how large is large enough?
- Using the Bonferroni correction, the cut off should be less than $t_{\alpha/2n, df=n-p-2}$ or greater than $t_{1-\alpha/2n, df=n-p-2}$ for $\alpha = 0.05$
- For reasonable size n , $|t_i| \geq 3$ will be considered an outlier;
- For $2 \leq |t_i| \leq 3$, it maybe suspicious depending on α and n ;
- Many high $|t_i|$'s indicates inadequate model.

```
library(faraway)
data(savings, package="faraway")
lmod <- lm(sr ~ pop15 + pop75 + dpi + ddpi, savings)
studres <- rstudent(lmod) # studentized residuals
studres[which.max(abs(studres))]
Zambia
2.853558
range(studres)
[1] -2.313429  2.853558
```

The studentized residuals may fail with multiple outliers

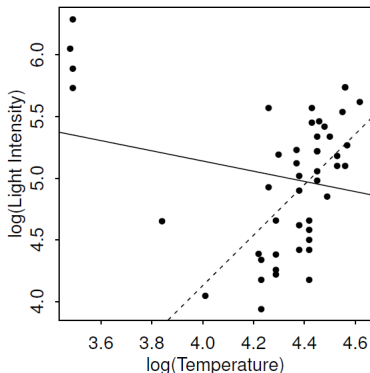


Figure 6.10 Regression line including four leftmost points is solid and excluding these points is dotted.

- The range of studentized residuals is $[-2.0494, 1.9058]$ (page 89 in the textbook);
- Based on the rule, there is **no outliers**!
- Studentized residuals have failed to identify outliers!
- Reason: **those outliers are grouped together**! (you need to keep this in mind when using studentized residuals to identify outliers.)

Identify influential points: Cook's distance

- An influential point is one whose removal from the dataset would cause a large change in the model fit.
- An influential point may or may not be an outlier and may or may not have large leverage;
- But an influential point will tend to have at least one of above two properties (being an outlier or have a large leverage).
- In other words, to identify influential points, we need to evaluate **how much change in model fit can be caused by removing one point**
- The **Cook's distance** is such a measure, defined as

$$D_i = \frac{(\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_{(i)})^T (\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_{(i)})}{(p+1) \times s^2} = \frac{1}{p+1} r_i^2 \frac{h_i}{1-h_i},$$

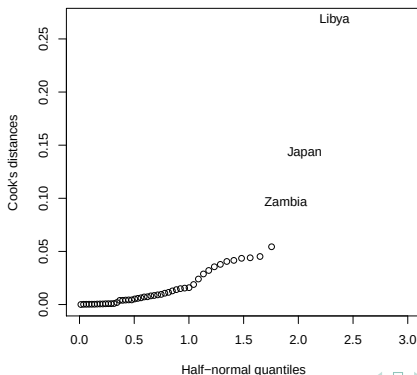
where p is the number of predictors in the model, $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$, $\hat{\mathbf{Y}}_{(i)} = \mathbf{X}\hat{\boldsymbol{\beta}}_{(i)}$, $s^2 = \frac{1}{n-p-1} \hat{\boldsymbol{\varepsilon}}^T \hat{\boldsymbol{\varepsilon}}$ and r_i 's are **standardized residuals** (3rd appearance!)

- r_i^2 : residual effect; h_i leverage; The combination of the two leads to influence measured by D_i 's.

Using R to calculate Cook's distance

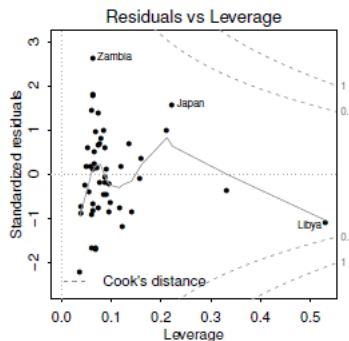
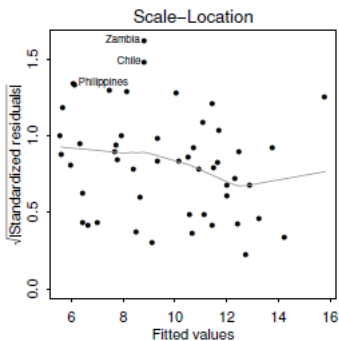
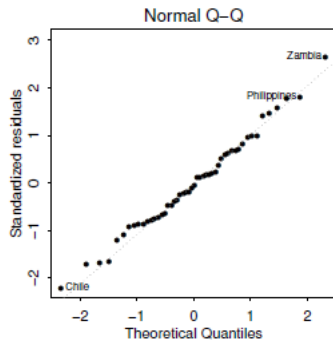
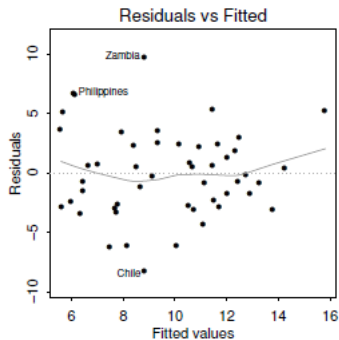
- The R command for Cook's distance: `cooks.distance()`

```
library(faraway)
data(savings,package="faraway")
lmod <- lm(sr ~ pop15 + pop75 + dpi + ddpi, savings)
cook <- cooks.distance(lmod)#find Cook's distance
#half-normal plot of Cook's distance
halfnorm(cook,3,labs=countries,ylab="Cook's distances",xlim=c(0,3))
```



Diagnostic plots for regression in R

```
library(faraway)
data(savings, package="faraway")
lmod <- lm(sr ~ pop15 + pop75 + dpi + ddpi, savings)
plot(lmod) # produce diagnostic plots
```



Inference of β_0

- $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = \sum Y_i/n - \sum \frac{\bar{X}(X_i - \bar{X})}{\sum (X_i - \bar{X})^2} Y_i$ is a function of $Y_1, Y_2, \dots, Y_n$
- $\hat{\beta}_0$ has mean β_0 . i.e. the least square estimate of β_0 is unbiased
- $\hat{\beta}_0$ has variance $[\frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2}] \sigma^2$
- $\hat{\beta}_0$ is UMVUE, i.e. unbiased and has minimum variance among all other unbiased linear estimator
- $\hat{\beta}_0 \sim N(\beta_0, [\frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2}] \sigma^2)$
- we use $[\frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2}] MSE$ to estimate the variance of $\hat{\beta}_0$
- $\frac{\hat{\beta}_0 - \beta_0}{s\{\hat{\beta}_0\}}$ follows what distribution?
t distribution with $n - 2$ df

Simple Linear Regression Model by Maximum Likelihood Function

- Suppose $x_1, \dots, x_n$, iid, is a sample such that each x_i has density $p(x_i | \theta)$.
- The likelihood function for this set of random variables is

$$\mathcal{L}(\theta) = \prod_{i=1}^n p(x_i | \theta). \quad (1)$$

- The likelihood function is considered a function of θ (the parameter of interest).
- The likelihood function measures how likely, or probable, it is to observe some value of θ given an observed sample of X .

- Given the definition of (1), it follows that a natural objective is to derive parameter estimates that maximize the likelihood function for any set of $x_1, \dots, x_n$.
- We call the estimate(s) described in the previous point, the **maximum likelihood estimate(s) (MLE(s))**.
- Formally, let $\mathcal{L}(\theta)$ be the likelihood function, for $\theta \in \Theta$. Then, a MLE is any value of $\hat{\theta} \in \Theta$ for which

$$\mathcal{L}(\theta) \leq \mathcal{L}(\hat{\theta})$$

- Note: rather than maximizing the likelihood function, it is often easier to work with the log-likelihood function.

- Recall: if $X \sim \mathcal{N}(\mu, \sigma^2)$, then the density function for X is

$$f(x | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x - \mu)^2}{2\sigma^2} \right\}.$$

- For the SLR Model, $\epsilon \sim \mathcal{N}(0, \sigma^2)$.

- Therefore, the density of ϵ is given by

$$f(\epsilon | \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{\epsilon^2}{2\sigma^2} \right\}.$$

- It follows that the likelihood function is given by

$$\mathcal{L}(\sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{\epsilon_i^2}{2\sigma^2} \right\}.$$

- Substituting $\epsilon_i = y_i - (\beta_0 + \beta_1 x_i)$ gives

$$\mathcal{L}(\beta_0, \beta_1, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2} \right\}. \quad (2)$$

- To maximize (2) with respect to the model parameters, we first find the log-likelihood function, $l(\beta_0, \beta_1, \sigma^2)$.
- The log likelihood function for a SLR model can be written as:

$$\begin{aligned} l(\beta_0, \beta_1, \sigma^2) &= \log \mathcal{L}(\beta_0, \beta_1, \sigma^2) \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2. \end{aligned}$$

- To get the MLEs for β_0 , β_1 , and σ^2 , we work out the score function for each parameter, e.g.,

$$S(\theta) = \frac{\partial l}{\partial \theta}$$

and then solve $S(\theta) = 0$ to get $\tilde{\theta}$.

- For β_0 , the score function is given by

$$S(\beta_0) = \frac{1}{\sigma^2} \left(\sum_{i=1}^n y_i - n\beta_0 - \beta_1 \sum_{i=1}^n x_i \right)$$

- Find the score functions for β_1 and σ^2 .

$$S(\beta_1) = \frac{1}{\sigma^2} \left(\sum_{i=1}^n x_i y_i - \beta_0 \sum_{i=1}^n x_i - \beta_1 \sum_{i=1}^n x_i^2 \right)$$

$$S(\sigma^2) = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

- Therefore, the MLE for β_0 is:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

- The MLEs for β_1 and σ^2 are:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n y_i (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}.$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2.$$

- Note: To ensure it is an MLE, the negative of the Hessian matrix must be positive definite. We will discuss this in more detail later